Printables
by JOSEF PRUSA

Minimal Case for BC250 and Flex PSU

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VIEW IN BROWSER

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Summary

A smallish, toolless case for the BC250 and a flex psu. Tested with an FSP500-30AS.

Gadgets > Computers

A simple, toolless case for the BC250. Who knows how bad the thermals are! Printed in PLA, it's probably fine. **The build assumes the center fins of the card have been opened up to accept a fan.**

BOM

1. BC250 APU Board
2. Flex PSU with right-angled PCIE connector
3. Standard 120mm case fan
4. (Optional) Female PCIE power socket

Print Instructions (SUPER IMPORTANT DO NOT SKIP)

The pieces are laid out, ready to print in the 3mf. If you want to muck with it, the two shells are meant to be printed on the flat ends, the harness pieces have a chamfer to place on the build plate and print at an angle. If you trust your overhangs, all pieces can be printed without supports. The IO Shield/Rear Harness has a small lip across the horizontal pieces for attaching a support otherwise. I print at 2x.6mm perimeters with 5% infill, but it should work at 3x .4mm.

The rear and front shells have a ‘mesh’ part, which should be printed with custom layers and infill:

- 0 top/bottom layers
- 3 (or 4, with a .4mm nozzle) perimeters
- 15% infill
- Rectilinear fill pattern

This creates the airflow mesh on the ends.

Print one each of the front shell, rear shell, front harness and rear harness.

Update v1.1

The case in the 3mf can get a little warm. A new shell has been added with extra ventilation on the sides, dimensions are unchanged and can use the existing harness. Refer to the 3mf for printing orientation.

Update v1.2

Chamfers were added to the shell to reduce sharp corners in printing, retaining block to added to front shell to keep psu from sliding inwards when pressed too hard. Fan screw hole sizing has been fixed, maybe. Who knows.

Assembly

1. Remove the APU's metal bracket
2. Slot the card into the two harness pieces
3. Slide the PSU into the retainers in the rear shell. It should be a snug friction fit.
4. Slide the card IO shield first into the rear shell, two small locator pins will click slightly when it is seated.
5. (Optional) Desolder the PCIE molex socket, replace it with a straight vertical one.(Optional) Desolder the PCIE molex socket, replace it with a straight vertical one.
6. Connect the right angle PCIE power supply (or the straight one if you replaced the socket), attach the 120mm fan header, and slide the fan between the board and the rear shell. Screw in two of the fan holes.
7. Slide the front shell onto the card. Keep in mind that the power cables only fit over the board, then between the gap created by the harness. Screw the remaining two fan screws.

Things Which Suck About the Case

Good luck opening it without the whole thing spilling out! Also, the front and back mesh really should have been pop-in but I got lazy. The holes for the fan screws also suck. Probably I should design a mesh to go over the fan so you don't chop your fingers off when it's running, but I would like to take a nap. I printed this exactly once, it's possible that the thermals are real crummy. There's space for adding a passive heat sink to the bottom of the board, and also probably to just throw a second case fan in there. Maybe the sides should also be vented?! We'll find out! Also, it could really use a place somewhere to add a switch to the power supply, because these FSPs have coil whine when not loaded, but I didn't do it! So... bummer! Anyway, if you want to be the one to fix these things, please do!

https://cad.onshape.com/documents/c3e8b3a5eff4e94305268ca5/w/d9a81dd26a493e4cc3e2e78a/e/935991319d53c60f5e5fcd46

Model files

v1.1 2 files

bc250-case-front-shell-v11.stl

bc250-case-rear-shell-v11.stl

v1.2 2 files

bc250-case-front-shell-v12.stl

bc250-case-rear-shell-v12.stl

STEP - Full

1 file

bc250-case-step-full.step

STEP - Parts

7 files

bc250-case-rear-shell-mesh.step

bc250-case-rear-shell.step

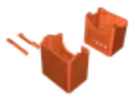
bc250-case-front-harness.step

bc250-case-front-shell.step

bc250-case-rear-shell-io-box.step

bc250-case-front-shell-mesh.step

bc250-case-rear-harness.step



bc250-case-three-plates.3mf